

Electro Mechanical Systems (EE2010)

Date: April 4th, 2024

Course Instructor(s)

Mr. Muhammad Abdul Majid

Mr. Syed Muhammad Ismaeel

Sessional-2 Exam

Total Time: 1 Hours

Total Marks: 50

Total Questions: 02

Semester: SP-2024

Campus: Lahore

Dept: Electrical

Engineering

Student Name

Roll No

Section

Student Signature

Vetted by

Vetter Signature

CLO #3: Analyze Synchronous Generator performance along with special emphasis towards environmental cost of generation

Q1:

[marks 25]

A 2.6 KV, 1.25 MVA, 0.85-PF-lagging, 50-Hz, 16 pole, Y-connected synchronous generator, has a synchronous reactance of 1.2Ω and an armature resistance of 0.2Ω . At 50 Hz, its friction and windage losses are 26 KW, and its core losses are 22 KW. The field circuit has a dc voltage of 220 V, and the maximum I_F is 10 A. The resistance of the field circuit is adjustable over the range from 22 to 200 Ω .

The generator is running at rated conditions, find out the following quantities

- a) Line current
- b) Rotational speed
- c) Voltage regulation
- d) Efficiency
- e) Output torque of prime mover

$$\frac{V_{NL} - V_{FL}}{V_{FL}} \times 100$$

I_F (A)	2	4	6	8	10
$V_{T,NL}$ (V)	1200	2200	2700	2850	2950

OCC for above generator

CLO #2: Investigate working of a DC Machine

Q2:

[marks =25]

A 20KW, 240V compensated DC shunt motor has armature resistance of 0.4Ω . Its field resistance is 100Ω and adjustable resistance connected in series with field coil can vary from 50Ω to 200Ω . Its no load characteristic curve at a speed of 1500 rpm is tabulated as under.

I_F (A)	0.3	0.6	0.9	1.2	1.4	1.6
E_A (V)	130	230	274	288	292	295

Find out following quantities assuming rated terminal voltage in all cases.

- a) No load speed of motor if adjustable resistance connected in series with field coil has been set to a value of 110 ohm.
- b) The speed at a line current of 60A, with same field current as in part (a)
- c) Fully labeled Circuit diagram for part (b)
- d) Output torque in part (b) if mechanical and core losses are 700 watts
- e) Maximum possible no load speed of the motor

$$\uparrow \omega \propto \frac{1}{\downarrow}$$